Requantizer Microarchitecture

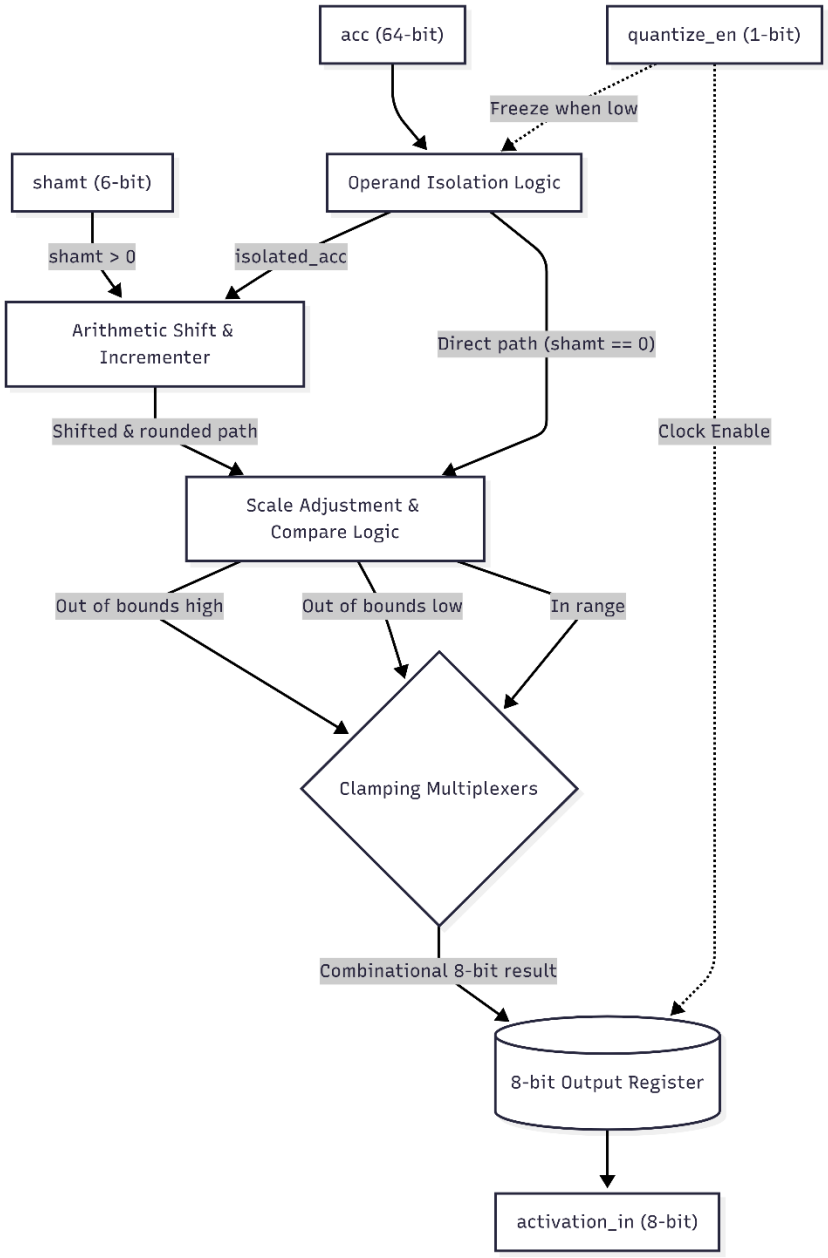

Figure 1. Block-level microarchitecture.

High-Level Overview

The Requantizer module serves as the quantization and scaling stage of the datapath. It takes the full-precision 64-bit output from the main arithmetic unit and scales it down into a saturated 8-bit signed integer.

Interface

Port Name	Direction	Width	Description
clk	Input	1 bit	Main system clock.
rst	Input	1 bit	Active-low asynchronous reset.
quantize_en	Input	1 bit	The trigger signal driven by the Controller when the SHIFT_RIGHT opcode is active AND the MAC unit's internal done signal is high. When high, it enables the internal combinational scaling logic and tells the module to register the scaled result on the next clock edge.
acc	Input	64-bit bus	The raw, signed 64-bit accumulation result from the MAC unit.
shamt	Input	6-bit bus	The dynamic shift amount used to scale the 64-bit value. This determines how many bits the data is arithmetically shifted to the right before rounding and clamping.
activation_in	Output	8-bit bus	The final, scaled, and saturated signed integer. This value is registered and serves as the direct input to the downstream Activation module.

Quantization Datapath

Operand Isolation (Power Optimization)

The module implements operand isolation to prevent unnecessary switching activity. The 64-bit acc input is routed through an isolation barrier implemented as a 2-to-1 multiplexer controlled by the quantize_en signal.

The output of this multiplexer generates a new internal signal called isolated_acc. When quantize_en is low, isolated_acc is forced to zero, keeping the downstream shifter and adders quiet while the MAC unit calculates. When quantize_en is asserted, isolated_acc passes the true accumulator value forward into the scaling logic.

Dynamic Shifting and Hardware Rounding

When the shift amount (shamt) is greater than zero, the module performs a round-half-up quantization to preserve precision. The hardware first performs an arithmetic right shift on the 64-bit isolated_acc by shamt - 1. The datapath then adds a constant 1 to this shifted value, generating a plus_one signal.

Saturation and Clamping Logic

The module enforces strict signed boundaries of -128 to +127. Before the bounds check, the hardware standardizes the data scale:

- **Direct Path (shamt == 0):** The evaluation uses the raw 64-bit isolated_acc.
- **Shifted Path (shamt > 0):** The hardware first slices the least significant bit off the rounded plus_one signal, effectively dividing it by 2, to bring it into its final integer scale.

The saturation multiplexers then evaluate this scale-adjusted 64-bit value against standard 8-bit boundaries. If the value is greater than 127, the output is clamped to 8'sd127. If the value is less than -128, the output is clamped to -8'sd128. If the value falls within the valid range, the lower 8 bits are truncated and routed to the output register.

Clocking and Critical Path

The Requantizer module consists of combinational logic followed by a synchronous output register. All sequential logic within the module operates synchronously on the positive edge of the main system clock (clk). The module utilizes an active-low asynchronous reset (rst), which immediately clears the 8-bit activation_in output register to zero. Normal operation is gated by the quantize_en control signal, which the output register treats as a synchronous clock enable, capturing the evaluated result only on the clock edge where it is asserted high.

While quantize_en controls the sequential capture, it also acts as the gate for the combinational arithmetic. The critical timing path spans through four strictly combinational logic stages:

- **Operand Isolation:** The signal must first pass through the isolation gating logic controlled by quantize_en.
- **Variable Shifter:** The data then passes through a 64-bit shifter to perform the arithmetic shift.
- **64-bit Incrementer:** The shifted result immediately feeds into a 64-bit adder to compute the rounding signal.
- **Scale Adjustment and Bounds-Check:** The rounded result is bit-sliced to restore the integer scale, and then routes into dedicated compare logic to evaluate the standard 8-bit saturation boundaries before final multiplexing.